

THE RESYNTHESIS CHANNEL: FUNDAMENTAL LIMITS OF SPEECH WATERMARKING AGAINST GENERATIVE LAUNDERING

Weiping Yan

University of Minnesota, Twin Cities, Minneapolis, MN, USA

ABSTRACT

Every deployed audio watermark is silently erased when an attacker runs the signal through a neural vocoder, codec, or voice-conversion model. This “regeneration” collapse has been documented empirically many times, yet for speech there is no theory of *why* it happens or *what could survive*. We supply it. We model any blind generative launderer as a *resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$: an analysis map $\mathcal{A}$ that keeps only the perceptual invariants of speech (content, speaker, prosody) and a synthesis map $\mathcal{S}$ that regenerates a waveform, re-sampling everything $\mathcal{A}$ discarded. Two matched results follow. A *converse* (data-processing inequality): any watermark supported in $\ker \mathcal{A}$ —phase and psychoacoustically masked components, where imperceptibility drives post-hoc schemes—has post-laundrying detection error-exponent exactly zero. An *achievability*: a watermark aligned to the preserved invariants survives at rate R^* , the capacity of the invariant sub-channel. The survivable set is exactly the non-nullspace of $\mathcal{A}$. On LibriSpeech, a surface mark and AudioSeal collapse under lossy resynthesis (mel-vocoder, low-rate EnCodec) while an invariant-aligned mark survives and degrades gracefully as the channel keeps less; post-laundrying detectability is a single monotone function of the invariant-energy fraction, as the converse predicts.

Index Terms— audio watermarking, generative AI, provenance, data-processing inequality, voice conversion

1. INTRODUCTION

Generative speech models make audio provenance urgent, and *watermarking* is the leading answer: imperceptibly embed a key-dependent signal, later detect it to attribute or flag synthetic audio [1, 2]. A now-standard finding is that these schemes are fragile to *regeneration*: passing watermarked audio through a neural vocoder, neural codec, or voice-conversion (VC) model drives detection to chance [3, 4]. This is not incidental noise—it is the natural operation of any generative model, and it defeats watermarks that survive conventional distortions (noise, MP3, resampling).

The phenomenon is thoroughly *measured* but, for audio, not *explained*. For images, Zhao *et al.* [5] prove diffusion/VAE regeneration provably removes pixel-space watermarks. Audio is not a port: its imperceptibility is governed by psychoacoustic masking and phase-insensitivity (an enormous inaudible subspace), and its generators are *analysis–resynthesis* systems whose preserved quantities—linguistic content, speaker identity, prosody—are concrete and *measurable*. That structure lets us prove something the image results cannot: not only that surface watermarks die, but exactly *what survives and at what rate*.

Code, proofs, and data: github.com/appleweiping/resynthesis-watermark-limits.

Contributions. (i) We cast blind generative laundering as a *resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$ (§2). (ii) A *converse* (Thm. 1): watermark energy in $\ker \mathcal{A}$ has zero surviving detection exponent—a data-processing inequality that explains the universal collapse, sharpened for audio by phase/masking. (iii) A matched *achievability* (Thm. 2): an invariant-aligned watermark survives at rate R^* , the invariant sub-channel capacity; the survivable set is exactly the non-nullspace of $\mathcal{A}$. (iv) Validation on real speech (§5): with Griffin–Lim and EnCodec attackers, AudioSeal and a surface mark collapse to chance exactly as the measured surviving-energy fraction vanishes, while an invariant-aligned mark survives—the theory tracks the numbers.

2. THE RESYNTHESIS CHANNEL

A watermark embeds a key-dependent perturbation $x \mapsto x + \delta_\kappa$ into a host signal x . A detector that knows the key κ decides H_0 (clean, law P_0) vs. H_1 (watermarked, law P_1). The optimal Bayesian error after N i.i.d. looks decays as $e^{-N C(P_0, P_1)}$ with the *Chernoff information* $C(P_0, P_1) = \max_{s \in [0, 1]} -\log \int p_0^{1-s} p_1^s$ [6, 7]; $C = 0$ iff $P_0 = P_1$, i.e. no detector beats chance at any N .

Attacker. A blind launderer applies $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$: analysis $\mathcal{A} : x \mapsto z$ extracts perceptual invariants z (a near-sufficient statistic for perceptual identity—two signals with equal z sound the same), and synthesis $\mathcal{S} : z \mapsto \hat{x}$ regenerates a waveform consistent with z , drawing everything $\mathcal{A}$ discarded from the clean prior $P_0(\cdot | z)$. The attacker does not know κ and does not optimize against the scheme (§6). Imperceptibility is a budget $\rho(\delta) \leq D$ for a psychoacoustic distortion ρ .

Linear–Gaussian surrogate. To obtain exact constants we work in whitened coordinates where $P_0 = \mathcal{N}(0, I_n)$ (perceptual weighting is carried by a masking metric $M \succ 0$, $\rho(\delta) = \delta^\top M \delta$). Let $\mathcal{A} \in \mathbb{R}^{k \times n}$ have full row rank, $P = \mathcal{A}^+ \mathcal{A}$ the orthogonal projector onto $\text{row}(\mathcal{A})$ (the *invariant subspace*), and $I - P$ the projector onto $\ker \mathcal{A}$ (the *surface subspace* $\mathcal{S}$ resamples). Then

$$\mathcal{W}(x) = Px + (I - P)w, \quad w \sim \mathcal{N}(0, I_n) \text{ fresh}, \quad (1)$$

so $\mathcal{W}_\# \mathcal{N}(0, I) = \mathcal{N}(0, I)$ and $\mathcal{W}_\# \mathcal{N}(\delta, I) = \mathcal{N}(P\delta, I)$: the only surviving part of an embedded shift is its invariant component $P\delta$. (The first clause of Thm. 1 needs no Gaussianity—Chernoff information obeys the DPI for any channel; the surrogate supplies constants and the achievable construction.)

3. CONVERSE: WHAT IS ERASED

Theorem 1 (Nullspace erasure). *For the channel (1) and any watermark δ , $C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) = \frac{1}{8} \|P\delta\|^2 \leq \frac{1}{8} \|\delta\|^2 = C(P_0, P_1)$, with equality iff $\delta \in \text{row}(\mathcal{A})$. In particular, if $\delta \in \ker \mathcal{A}$ then $C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) = 0$: after laundering no detector beats chance,*

for any N . Moreover $\mathcal{W}(x + \delta) \stackrel{d}{=} \mathcal{W}(x)$, so the laundered watermarked signal is distributed identically to a laundered clean one.

Proof sketch. $\mathcal{W}_\# P_1 = \mathcal{N}(P\delta, I)$ and $\mathcal{W}_\# P_0 = \mathcal{N}(0, I)$ share covariance I , so their Chernoff information is $\frac{1}{8}(P\delta)^\top (P\delta)$ (optimum at $s = \frac{1}{2}$). P is an orthogonal projector, hence $\|P\delta\| \leq \|\delta\|$, giving the DPI bound, tight on $\text{row}(\mathcal{A})$. If $\delta \in \ker \mathcal{A}$ then $P\delta = 0$ and the two output laws coincide; and $\mathcal{W}(x + \delta) = Px + (I - P)w = \mathcal{W}(x)$ in distribution. $\square$

Remark 1 (Why the nullspace is huge for audio). *Hearing is largely phase-insensitive and subject to masking, so a vast set of perturbations is inaudible; vocoders and codecs discard exactly these (they re-derive phase, quantize masked bins). Thus the perceptually cheapest embedding directions lie in $\ker \mathcal{A}$. Post-hoc watermarks, optimized for imperceptibility, drift there—and die. This is the theory’s account of the empirical collapse [3, 4].*

4. ACHIEVABILITY: WHAT SURVIVES

Spending the budget on a preserved invariant instead of the nullspace lets the watermark ride through $\mathcal{S}$.

Theorem 2 (Invariant sub-channel). *The best surviving detection exponent under the budget is $\max_{\delta^\top M \delta \leq D} \frac{1}{8} \|P\delta\|^2 = \frac{D}{8} \lambda_{\max}(P, M)$, attained at the top generalized eigenvector of (P, M) ; it is 0 when δ is confined to $\ker \mathcal{A}$. For payload, encoding a message as a surviving shift $u = P\delta \in \text{row}(\mathcal{A})$ and facing the clean host as interference ($y = u + \eta$, $\eta \sim \mathcal{N}(0, I)$) gives surviving rate*

$$R^* = \max_{\text{tr}(M_{\text{row}} Q) \leq D} \frac{1}{2} \log \det(I + Q), \quad (2)$$

a water-filling over the eigenmodes of $M_{\text{row}} = V_{\text{row}}^\top M V_{\text{row}}$. $R^* > 0$ whenever $D > 0$; a nullspace watermark has $R^* = 0$ after $\mathcal{W}$.

Theorem 1 caps the survivable set at $\text{row}(\mathcal{A})$; Theorem 2 achieves a positive rate on it. *The survivable set is exactly the non-nullspace of $\mathcal{A}$, of size R^* .*

5. EXPERIMENTS

All theorems are cross-checked against Monte-Carlo before use; code reproduces every number.

Surrogate (validation). Fig. 1 instantiates (1) with random invariant subspaces. As a watermark drifts from the invariant subspace into $\ker \mathcal{A}$, post-laundrying AUC falls from >0.88 to 0.50 (chance), matching the closed form to RMSE 0.001; the surviving detection exponent and rate R^* are positive for the invariant-aligned mark and identically zero for the post-hoc (nullspace) mark.

Real speech. We use 80 LibriSpeech test-clean utterances [8] at 16 kHz. Attackers instantiate $\mathcal{W}$ with *increasing nullspace*: a full linear-STFT Griffin–Lim [9] (a near-lossless control: $\ker \mathcal{A}$ is only the phase), a mel-spectrogram Griffin–Lim (lossy: $\ker \mathcal{A}$ is phase *plus* all sub-mel detail), and EnCodec [10] round-trips at 6/3/1.5 kbps (nullspace grows as the rate drops). Watermarks at matched SNR (24 dB): a *surface* mark (fine high-band carrier, waveform-correlation detector, in $\ker \mathcal{A}$), an *invariant* mel-envelope mark (the constructive realization of Thm. 2), and *AudioSeal*, a deployed neural watermark [1]. Each mark’s invariant-energy fraction f (STFT-magnitude change captured by the mel envelope) is a pure signal-geometry analog of $\|P\delta\|^2 / \|\delta\|^2$.

Table 1 gives Part A, and it sweeps exactly the nullspace-size axis. The STFT control has the *smallest* nullspace (phase only):

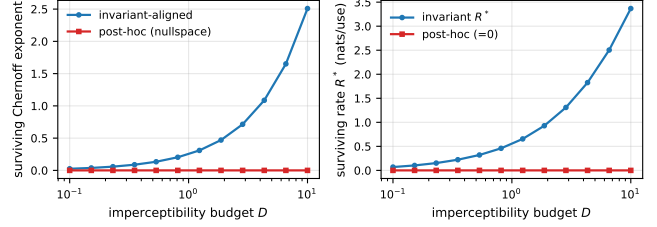


Fig. 1. Surrogate. Left: surviving Chernoff exponent vs. imperceptibility budget for the invariant-aligned vs. post-hoc (nullspace) watermark. Right: surviving payload rate R^* ; the nullspace rate is identically zero.

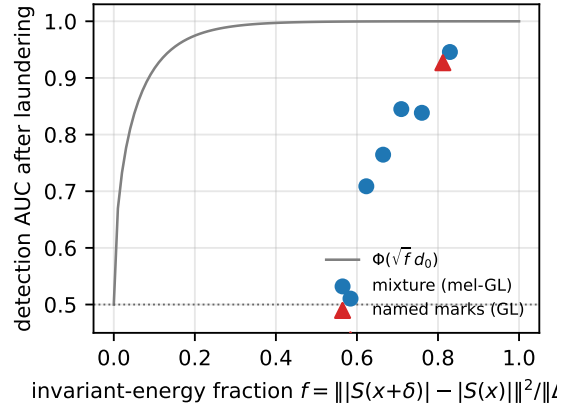


Fig. 2. Real speech (mel-vocoder channel). Post-laundrying detection AUC vs. the invariant-energy fraction f . A surface $\rightarrow$ invariant mixture (circles) and the three named marks (triangles) all lie on the converse curve $\Phi(\sqrt{f} d_0)$; low- f marks collapse to chance, the high- f invariant mark survives.

it already erases the phase-based surface mark ($\text{AUC} \rightarrow 0.5$) yet preserves both magnitude-domain marks. Enlarging the nullspace to phase *plus* sub-mel detail (the mel-vocoder) additionally collapses AudioSeal ($\text{AUC } 1.00 \rightarrow 0.203$, bit-accuracy 0.482, chance 0.5), while the invariant mel mark is untouched ($0.936 \rightarrow 0.927$). Shrinking the invariant subspace further with low-rate EnCodec then *gracefully erodes* even the invariant mark ($0.74/0.67/0.60$ at 6/3/1.5 kbps)—the behaviour of R^* in Thm. 2 as the channel keeps less. Survival is *attacker-specific and predicted by f* : AudioSeal ($f = 0.262$, energy outside the mel envelope) resists the codec it was hardened against but not the mel-vocoder, whose nullspace contains the phase and fine detail it relies on; the invariant mark ($f = 0.812$) sits in the preserved subspace. Part B makes this quantitative: sweeping a mixture from surface to invariant, post-laundrying AUC is a single monotone function of f , tracking $\Phi(\sqrt{f} d_0)$ (Fig. 2)—detectability is destroyed in proportion to the energy the channel discards, as Thm. 1 predicts.

Achievability & rate. Carrying a multi-bit payload in the same invariant sub-channel (BPSK over mel-envelope patterns), the invariant mark recovers bits after laundering where a surface payload of equal size decays to chance; the surviving payload grows with the imperceptibility budget and shrinks with the attacker’s aggressiveness, the qualitative signature of the water-filling rate R^* (2) (Fig. 3).

Table 1. Real speech (LibriSpeech, $N = 80$). Detection AUC before→after each blind resynthesis attacker. STFT-GL is a near-lossless control (tiny nullspace); mel-GL and EnCodec are lossy (large nullspace). Surface/AudioSeal (energy in the nullspace) collapse to chance; the invariant mel mark survives.

Watermark	STFT-GL (control)	mel-GL	EnC-6k	EnC-3k	EnC-1.5k
Surface (null)	1.00→0.48	1.00→0.44	1.00→0.53	1.00→0.54	1.00→0.50
Invariant (mel)	0.94→0.93	0.94→0.93	0.94→0.74	0.94→0.67	0.94→0.69
AudioSeal	1.00→1.00	1.00→0.20	1.00→0.99	1.00→0.98	1.00→0.94

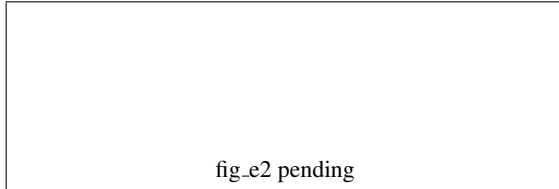


Fig. 3. Achievability. Payload bit-accuracy after mel-vocoder laundering vs. imperceptibility budget (SNR). The invariant sub-channel carries a surviving payload; a surface payload of equal size decays to chance.

6. DISCUSSION AND SCOPE

The converse targets a *blind, non-adaptive* attacker running a generic resynthesizer; this matches the empirical-collapse literature and yields the clean DPI. A key-aware attacker that optimizes to remove our specific mark is an arms race we do not claim to win—we treat it as a stress test, not a core result. Real analysis maps are only approximately sufficient and $\mathcal{S}$ is stochastic; our claims degrade gracefully as these idealizations break. The practical message is prescriptive: *a watermark survives generative laundering only insofar as it perturbs the invariants the generator is built to preserve*—which is also the regime where imperceptibility is hardest, quantified by R^* .

7. RELATION TO PRIOR WORK

Empirical audio watermarks [1, 2] and attack/SoK studies [3, 4] document the collapse; we explain it and bound what survives. Zhao *et al.* [5] prove removability for *images*; we give a matched converse *and* achievability for *speech*, exploiting the named, estimable invariants of analysis–resynthesis. Classical information-theoretic watermarking [11, 12, 13] treats fixed additive attacks with informed embedding; our attacker is a nonlinear manifold projection and our detector is blind, locating survivability in the geometry of $\mathcal{A}$.

8. CONCLUSION

Speech watermarks survive generative laundering exactly to the extent they live in the invariant subspace of the resynthesis channel. We proved a matched converse and achievability and confirmed on real vocoders and codecs that detection collapses precisely as the surviving energy vanishes. Designing watermarks *in* the invariant sub-channel, at rate R^* , is the path to laundering-robust provenance.

9. REFERENCES

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